VERIHIRE

AI-POWERED SKILL CERTIFICATION AND HIRING PLATFORM

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INTRODUCTION

- Traditional resumes and self-claimed skills lack credibility.
- Recruiters face hiring risks due to resume fraud and unverifiable claims.
- Skill certifications today are often theoretical and do not reflect real-world ability.
- There is a growing need for a transparent, merit-based system where skills are validated practically.

PROBLEM STATEMENT

To develop a platform that combats resume frauds by using AI-generated skill challenges with automated scoring and peer validation, creating verified portfolios for candidates and giving recruiters a transparent, skill-based hiring dashboard.

OBJECTIVES

- Verify Real-World Skills – Use AI-generated challenges to test practical abilities beyond resumes.
- Automate Skill Evaluation – Apply AI models for unbiased, consistent scoring.
- Build Dynamic Portfolios – Provide candidates with auto-updated, skill-based profiles.
- Streamline Hiring – Offer recruiters a dashboard to filter and shortlist based on verified skills.

KEY CHALLENGES

- Evaluating AI models on real-world vocational skills is limited.
- Lack of trust and transparency in AI certification processes.
- Peer grading suffers from bias, collusion, and low-effort reviews.
- Digital certificates are vulnerable to forgery and verification issues.

EXISISTING SOLUTIONS

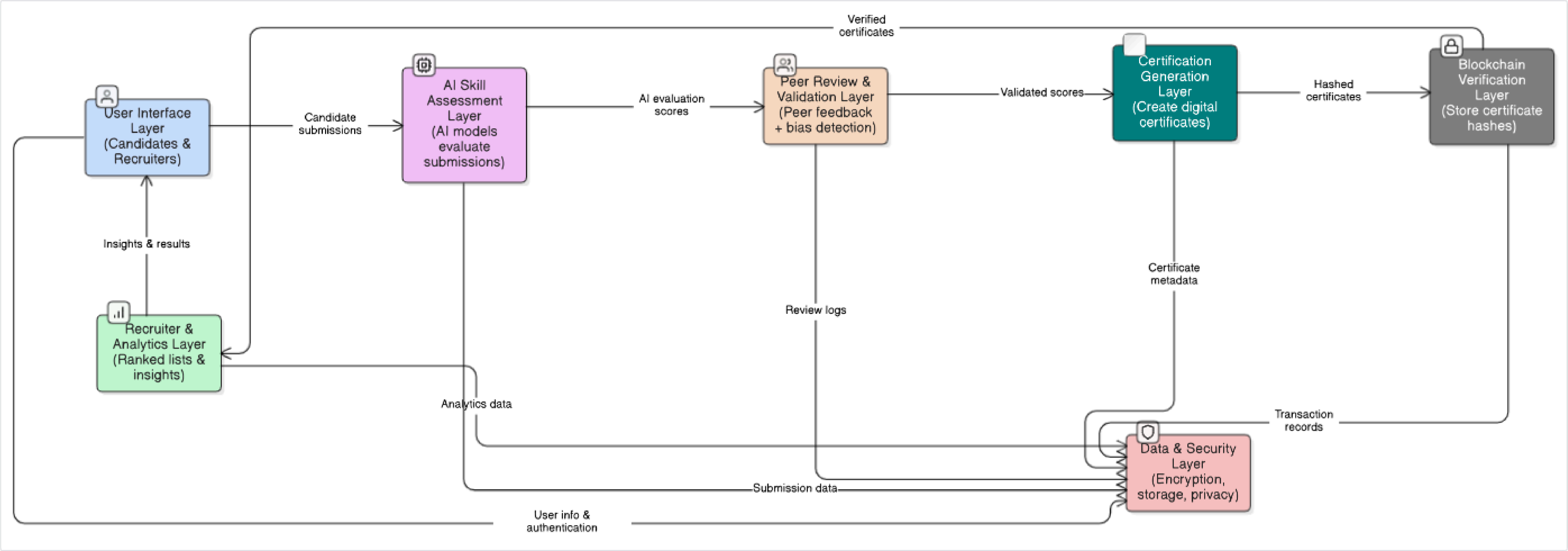
- Benchmark datasets (e.g., Professional Certification Benchmark) for AI exam-style testing.
- Governance frameworks (e.g., AI Certification by Cihon et al.) to reduce info asymmetry.
- Blockchain-based platforms (e.g., SkillCheck) for securing the certficates.
- Peer-grading systems (e.g., CrowdGrader) with reputation-weighted scoring.
- Blockchain credential standards (e.g., Blockcerts) for tamper-proof certificates

LITERATURE REVIEW

Reference	Methodology	Drawbacks
D. Noever and M. Ciolino, "Professional Certification Benchmark Dataset: The First 500 Jobs for Large Language Models," PeopleTec, Inc., Huntsville, AL, USA, Tech. Rep., May 2023.	Built benchmark dataset of certification/exam items; evaluated LLMs (GPT-3/3.5) on vocational readiness using quantitative scoring	Focuses on LLM QA performance; doesn't reflect real-world hands-on skill certification
BP. Cihon, M. Kleinaltenkamp, J. Schuett, and S. D. Baum, "AI Certification: Advancing Ethical Practice by Reducing Information Asymmetries," Global Catastrophic Risk Institute (GCRI), Working Paper 21-1, 2021	Conceptual / policy analysis; proposes AI certification principles and governance pathways	High-level guidance; lacks concrete technical implementation or automated scoring

J. Gupta and S. Nath, “SkillCheck: An Incentive-based Certification System using Blockchains,” Indian Institute of Technology Kanpur, Technical Report, 2020.	Certificates are issued digitally and anchored on blockchain for secure, tamper-proof verification. Learners and evaluators are incentivized with rewards to motivate participation.	Requires proper blockchain infrastructure for deployment. Assessment quality depends on platform design; blockchain ensures security but not validity
MIT Media Lab and Learning Machine, “Blockcerts: An Open Infrastructure for Academic Credentials on the Blockchain,” Whitepaper, 2016.	Blockchain-based verifiable credential standard; defines issuing, storing, verifying workflows	Secures certificates but not assessment quality; privacy, interoperability, and cost issues
L. de Alfaro and M. Shavlovsky, “CrowdGrader: Crowdsourcing the Evaluation of Homework Assignments,” in Proceedings of the 2013 ACM Conference on Learning at Scale, San Jose, CA, USA,pp. 1–10.2013	Peer-grading system for assignments; iterative reputation-weighted aggregation; field experiments	Reviewer bias, collusion, or low-effort reviews may reduce reliability; depends on good incentives

ARCHITECTURE


Verified certificates
User Interface Layer (Candidates & Recruiters)
Candidate submissions
AI Skill Assessment Layer (AI models evaluate submissions)
AI evaluation scores
Peer Review & Validation Layer (Peer feedback + bias detection)
Validated scores
Certification Generation Layer (Create digital certificates)
Hashed certificates
Blockchain Verification Layer (Store certificate hashes)
Insights & results
Recruiter & Analytics Layer (Ranked lists & insights)
Review logs
Certificate metadata
Analytics data
Transaction records
Data & Security Layer (Encryption, storage, privacy)
Submission data
User info & authentication

1. User Interface Layer

Input:

- Candidate account info, selected skill domain (coding/design)
- Job role filters and recruiter preferences

Process:

- Provides submission portal for candidates to upload code/design tasks.
- Displays real-time progress of AI evaluations and peer review feedback.
- Recruiters access the dashboard, filter candidates, and view analytics.

Output:

- Candidate submissions forwarded to AI Skill Assessment Layer.
- Recruiters receive visual insights and ranked candidate lists.

2. AI Skill Assessment Layer

Input:

- Candidate submissions (code, design, written tasks)
- Candidate profile & skill level

Process:

- Challenge Generation (GPT-5): produces dynamic, real-world tasks based on profile & job role.
- Automated Evaluation:
  - CodeBERT scores coding logic and correctness.
  - ViT scores design submissions for creativity and quality.
  - BERT analyzes textual responses.
- Scores are normalized and feedback is generated automatically.

Output:

- Initial candidate scores and AI-generated feedback sent to Peer Review Layer.

3. Peer Review & Validation Layer

Input:

- Candidate submissions and AI-generated evaluation scores

Process:

- Peers review each other’s submissions anonymously.
- BERT models check for review quality and bias.
- Autoencoder / Anomaly Detection identifies irregular or collusive peer feedback.
- Combines AI and validated peer scores into a final verified score.

Output:

- Peer-verified, trustworthy candidate scores forwarded to Certification Layer.

4. Certification Generation Layer

Input:

Verified candidate scores and identity data.

Process:

- Generates digital certificates containing candidate details, skill data, and evaluation scores.
- Applies a SHA-256 hash to each certificate record for authenticity.
- Prepares metadata (timestamp, candidate ID, verification info).

Output:

Hashed certificate records forwarded to the Blockchain Verification Layer.

5. Blockchain Verification Layer

Input:

Hashed certificate data and metadata from the Certification Generation Layer.

Process:

- Stores the certificate hash and metadata on a blockchain or cryptographic ledger.
- Links the blockchain transaction ID to the certificate.
- Generates a QR code or verification portal link for easy authenticity checks.

Output:

Tamper-proof, blockchain-anchored skill certificates returned to both the candidate and the recruiter.

6. Recruiter & Analytics Layer

Input:

Verified candidate certificates and performance data.

Process:

- Uses Neural Collaborative Filtering (NCF) to rank candidates by skill-job fit.
- Dashboard visualizes analytics: top performers, skill trends, percentile ranks.
- Allows recruiters to filter by skills, experience, and certification validity.

Output:

Ranked candidate lists and actionable insights for recruiters.

7. Data & Security Layer

Input:

Data from all other layers (submissions, reviews, certificates, user information).

Process:

- Encrypts sensitive data using AES-256.
- Stores large files securely in IPFS or cloud storage.
- Manages authentication, authorization.

Output:

Secure, encrypted, and traceable data flow maintained throughout the system.

CONCLUSION

Verihire offers a transparent, AI-powered platform that validates real-world skills through automated evaluation and peer review.

It reduces hiring risks, combats resume fraud, and empowers both candidates and recruiters in building a merit-based ecosystem

REFERENCES

[1] W. Suktam and S. Lapchit, "Blockchain in education: Transforming learning, credentialing, and academic data management," Journal of Education and Learning Reviews, vol. 1, no. 6, pp. 37–46, 2024.

[[2] D. Gamage, T. Staubitz, and M. Whiting, "Peer assessment in MOOCs: A systematic literature review," Distance Education, vol. 42, no. 2, pp. 268–289, 2021.

[[3] P. Cihon, M. J. Kleinaltenkamp, J. Schuett, and S. D. Baum, "AI certification: Advancing ethical practice by reducing information asymmetries," IEEE Transactions on Technology and Society, vol. 2, no. 4, pp. 200–209, 2021.

[4] M. Jirgensons and J. Kapenieks, "Blockchain and the future of digital learning credential assessment and management," Journal of Teacher Education for Sustainability, vol. 20, no. 1, pp. 145–156, 2018.

[5] H. K. Suen, "Peer assessment for massive open online courses (MOOCs)," The International Review of Research in Open and Distributed Learning, vol. 15, no. 3, pp. 312–327, 2014.

Thank You